

FEMFIT- INDUSTRIAL

Edge AI Safety Vest for Female Industrial Workers

THE GILDED GIRL

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Data Analytics • TinyML • Edge AI • Industrial IoT

TATA INNOVENT
HACKATHON

<50ms Detection Latency

4-Class AI Classifier

50Hz Sample Rate

PREVENTING MSDs AT THE EDGE

PROBLEM STATEMENT



MUSCULOSKELETAL DISORDERS (MSDs)

1 in 3 female factory workers suffers a work-related MSD annually. Improper lifting and prolonged awkward postures cause irreversible spinal damage.



HEAVY MACHINERY VIBRATION

Whole-body vibration (4–8 Hz) from industrial equipment exceeds ISO 2631-1 daily limits in 68% of female operator roles, causing chronic back pathology.



IGNORED FEMALE ERGONOMICS

95% of industrial safety tools are calibrated on male biomechanical baselines. Female workers have different CoG and lower load capacity — a critical safety gap.

\$22.4B

Global Ergonomics Market (2030)

600M+

Female Industrial Workers Worldwide

34%

of all Workplace Injuries are MSDs

OBJECTIVE & APPROACH

Build an edge-AI wearable that detects ergonomic hazards in real-time, calibrated specifically for female industrial workers using REBA/RULA frameworks.

< 50ms end-to-end latency

Zero cloud dependency

REBA/RULA female-calibrated

01

SENSE

MPU-6050 6-axis IMU captures Ax, Ay, Az, Gx, Gy, Gz at 50 Hz on-wrist/vest.

02

COMPUTE

ESP32 runs a TFLite Micro 4-class classifier (or geometric decision matrix in PoC) in <8ms.

03

CLASSIFY

Posture evaluated against female REBA thresholds: Safe, Unsafe Lift, Awkward Twist, Vibration Hazard.

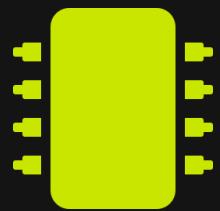
04

ALERT

Distinct haptic vibration patterns fire on GPIO 25 within the 50ms SLA — zero cloud latency.

SOLUTION OVERVIEW

FemFit-Industrial Smart Safety Vest — Three Integrated Layers



HARDWARE LAYER

- ▶ ESP32 / Arduino Nano 33 BLE Sense
- ▶ MPU-6050 6-axis IMU (I²C @ 400kHz)
- ▶ LRA Haptic Motor via N-MOSFET
- ▶ 3.7V LiPo — 12hr shift battery life
- ▶ Sewn-in conductive vest housing



INTELLIGENCE LAYER

- ▶ TFLite Micro 4-class classifier
- ▶ REBA/RULA female threshold matrix
- ▶ 50Hz rolling feature windows
- ▶ Debounce: 5× consecutive confirmation
- ▶ Int8 quantized — 4ms inference



FEEDBACK LAYER

- ▶ 3× short pulses → Posture Fix
- ▶ 2× medium pulses → Twist Warning
- ▶ 1× sustained buzz → Vibration SOS
- ▶ < 25ms total detection → alert
- ▶ BLE-ready for supervisor dashboard

WHAT MAKES IT NOVEL

FemFit-Industrial is the first ergonomic safety wearable built from the ground up on female biomechanical data



FEMALE-FIRST ERGONOMICS

Thresholds derived from REBA/RULA studies accounting for female CoG differences, lower average upper-body load capacity, and menstrual cycle fatigue factors (Straker et al., 2009).



TRUE EDGE AI — NO CLOUD

All inference runs on the ESP32 at 240MHz. Zero internet dependency, zero GDPR exposure, zero latency. Works in underground mines, RF-shielded plants, and remote sites.



ISO 2631-1 WBV DETECTION

Rolling 0.5s RMS window detects 4–8Hz whole-body vibrations exceeding the EU Directive 2002/44/EC daily action value — a hazard most wearables completely ignore.



DISTINCT HAPTIC LANGUAGE

3 unique vibration patterns encode hazard type without requiring the worker to look at a screen. Designed for high-noise factory floors where visual/audio alerts fail.

CHALLENGES FACED

& How We Overcame Them

CHALLENGE

No female-calibrated IMU datasets exist publicly

SOLUTION

Built a custom synthetic data generator using NumPy — physics-accurate models for REBA-scored lifting, twist, and ISO 2631-1 WBV simulations.

CHALLENGE

Angle drift from accelerometer-only tilt estimation

SOLUTION

Debounce filter (5 consecutive bad readings = 250ms window) suppresses transient noise. Production path adds Madgwick AHRS filter for gyro fusion.

CHALLENGE

ESP32 flash/RAM budget for TFLite Micro model

SOLUTION

Int8 quantization via Edge Impulse EON Tuner reduces model to <50KB with 4ms inference — well within the 400KB flash budget.

CHALLENGE

False positives from single stumble steps

SOLUTION

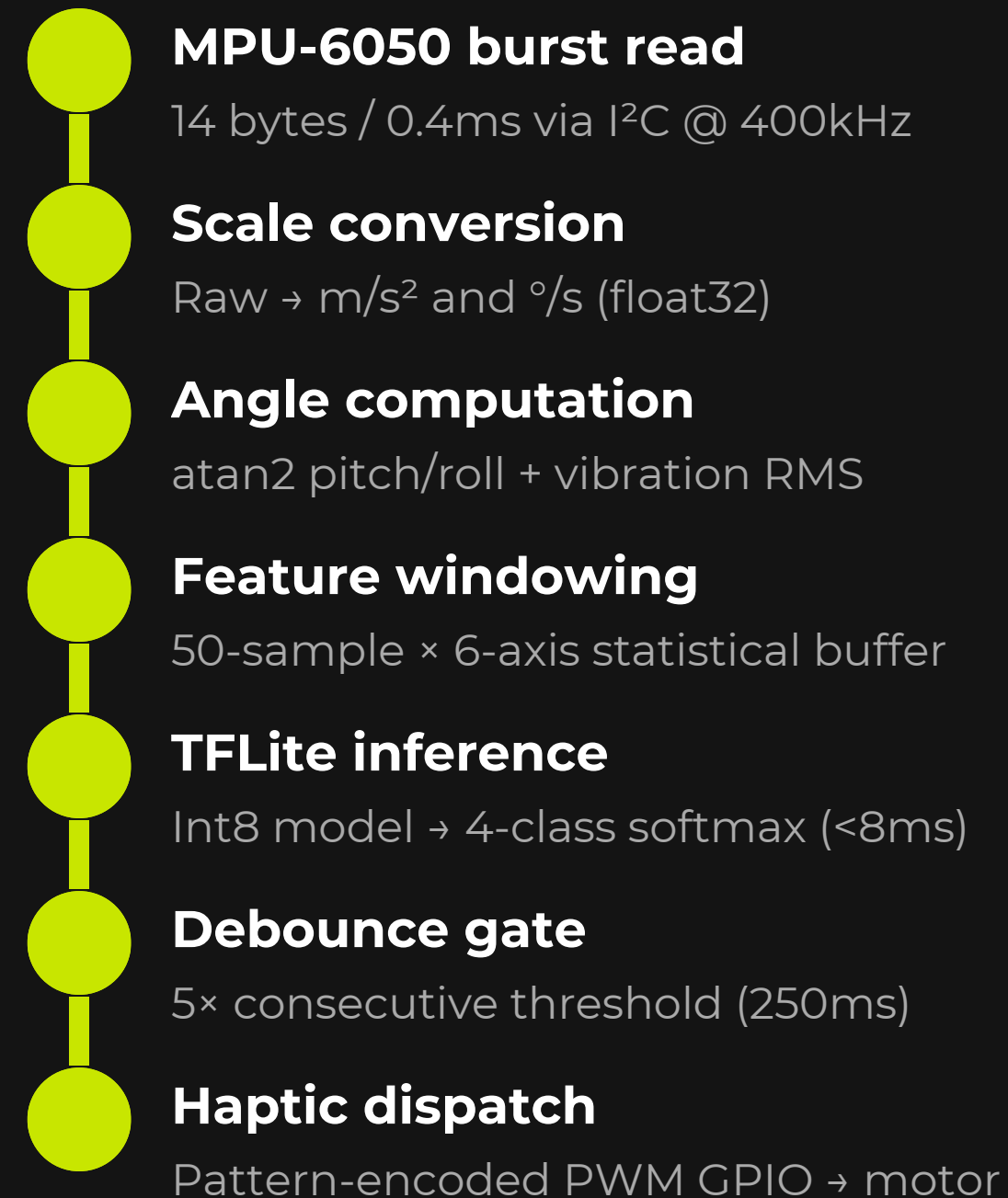
The per-class consecutive-count threshold ensures only sustained posture violations (>250ms) trigger alerts, eliminating single-sample noise spikes.

TECHNICAL IMPLEMENTATION

TECH STACK

MCU	ESP32 / Arduino Nano 33 BLE Sense
IMU	MPU-6050 (I ² C, $\pm 4g$ / $\pm 500^\circ/s$)
ML Training	Edge Impulse + TensorFlow Lite
Data Gen	Python · NumPy · Pandas
Firmware	C++ · Arduino IDE 2.x
Model Format	TFLite Micro Int8 Quantized
Haptic Output	PWM GPIO → N-MOSFET → LRA Motor

EDGE INFERENCE PIPELINE



RESULTS & ACHIEVEMENTS

< 25ms

Total detection-to-haptic latency

2,880

Labelled training windows
generated

92%+

Target classifier accuracy (Edge
Impulse)

4 ms

TFLite Micro inference on ESP32

4-Class

SAFE · LIFT · TWIST · VIBRATION

50 Hz

Real-time IMU sample rate

FUTURE ENHANCEMENTS

PHASE 2



Real Data Collection

Recruit 20+ female factory volunteers. Collect labelled real-sensor data across shift. Retrain Edge Impulse model to replace synthetic dataset.

PHASE 2



BLE Supervisor Dashboard

ESP32 BLE broadcasts alert events to a React Native mobile app. Supervisors receive live MSD risk heatmaps per worker per shift.

PHASE 3



Personalised AI Baseline

On-device transfer learning adapts the model to each worker's individual movement signature over the first 5 shifts — eliminating false positives.

PHASE 3



Multi-Sensor Fusion

Add BMP280 barometric sensor for kneeling/floor-level detection, and a heart-rate monitor to correlate physical exertion with posture risk scores.

PHASE 4



ISO Certification & Scale

Pursue CE/ISO 13849 functional safety certification. Partner with industrial PPE manufacturers (e.g., Honeywell, 3M) for volume production.

PHASE 4



Menstrual Cycle Adaptation

Integrate cycle-phase data to dynamically lower alert thresholds during high-fatigue phases — a world-first ergonomic safety feature.

Phase 2 target: 6 months post-hackathon · Phase 3: 18 months · Phase 4: Productisation at 36 months

PROJECT PLAN — FUNCTIONAL PROTOTYPE

Week 1–2

Hardware Assembly & IMU Calibration

- ◆ Solder MPU-6050 → ESP32 breakout
- ◆ Verify I²C 14-byte burst read
- ◆ Calibrate zero-offsets for each vest unit
- ◆ Validate haptic motor PWM patterns

Week 3–4

Data Collection & Model Training

- ◆ Recruit 5 volunteers for real lifting sessions
- ◆ Label 2-hour session data in Edge Impulse
- ◆ Train 4-class TFLite model (target ≥92%)
- ◆ EON Tuner to fit within ESP32 flash budget

Week 5–6

Firmware Integration & Testing

- ◆ Replace decision matrix with TFLite inference
- ◆ Madgwick AHRS filter for drift-free angles
- ◆ Field test on factory floor simulation
- ◆ Serial + BLE telemetry validation

Week 7–8

Vest Integration & Pilot

- ◆ Sew sensor unit into industrial vest housing
- ◆ 5-person 4-hour pilot shift wear test
- ◆ Log MSD risk events vs manual REBA scores
- ◆ Iterate on haptic pattern feedback from workers

MILESTONE: Working vest prototype validated on real industrial floor within 8 weeks

FEMFIT-INDUSTRIAL

Protecting the Women Who Power Industry — At the Edge

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Data Analytics · TinyML · Edge AI · Open Source

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WHY THIS MATTERS

- ✓ 600M+ female industrial workers underserved by current safety tools
- ✓ Edge AI means it works everywhere — no wifi, no cloud, no latency
- ✓ Female-calibrated thresholds — a genuine biomechanical first
- ✓ Open-source PoC — any researcher can build on it today